北京航空航天大學
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工科数学分析

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§15.2 多变量函数的求导

一、链式法则（ Chain Rule ）

定理 2.1: 如果函数$u=\phi(t)$及$v=\psi(t)$都在点t可导，函数$z=f(u,v)$在对应点(u,v)可微，则复合函数$z=f[\phi(t),\psi(t)]$在对应点 t 可导，且其导数可用下列公式计算：

$$\frac{dz}{dt}=\frac{\partial z}{\partial u}\frac{du}{dt}+\frac{\partial z}{\partial v}\frac{dv}{dt}.$$

证: 设 t 获得增量Δt，

由于函数$z=f(u,v)$在点(u,v)可微，

$$\therefore \Delta z=\frac{\partial z}{\partial u}\Delta u+\frac{\partial z}{\partial v}\Delta v+\alpha(\Delta u,\Delta v)\sqrt{(\Delta u)^2+(\Delta v)^2}$$

当$\Delta u\to 0$，$\Delta v\to 0$时，$\alpha(\Delta u,\Delta v)\to 0.$

$$\therefore \frac{\Delta z}{\Delta t}=\frac{\partial z}{\partial u}\cdot\frac{\Delta u}{\Delta t}+\frac{\partial z}{\partial v}\cdot\frac{\Delta v}{\Delta t}$$

$$+\frac{\alpha(\Delta u,\Delta v)\sqrt{(\Delta u)^2+(\Delta v)^2}}{\Delta t}$$

当$\Delta t \to 0$时， $\Delta u \to 0$，$\Delta v \to 0$

$$\frac{\Delta u}{\Delta t} \to \frac{du}{dt}, \quad \frac{\Delta v}{\Delta t} \to \frac{dv}{dt},$$

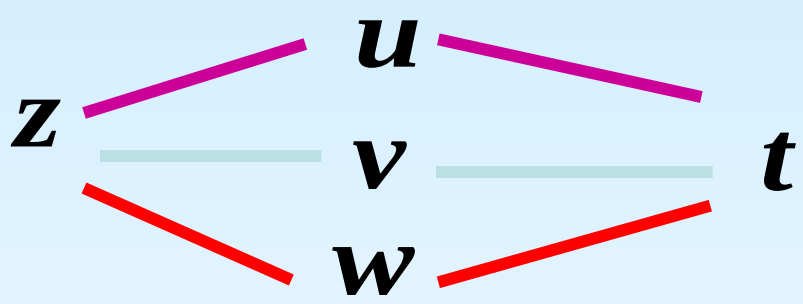
而 $$\lim_{\Delta t \to 0} \frac{\alpha(\Delta u, \Delta v)\sqrt{(\Delta u)^2 + (\Delta v)^2}}{\Delta t}$$

$$= \lim_{\Delta t \to 0} \frac{\alpha(\Delta u, \Delta v)\,|\Delta t|\sqrt{(\frac{\Delta u}{\Delta t})^2 + (\frac{\Delta v}{\Delta t})^2}}{\Delta t} = 0,$$

$$\therefore \frac{dz}{dt} = \lim_{\Delta t \to 0} \frac{\Delta z}{\Delta t} = \frac{\partial z}{\partial u} \cdot \frac{du}{dt} + \frac{\partial z}{\partial v} \cdot \frac{dv}{dt}.$$

上定理的结论可推广到中间变量多于两个的情况.

如



$$\frac{dz}{dt}=\frac{\partial z}{\partial u}\frac{du}{dt}+\frac{\partial z}{\partial v}\frac{dv}{dt}+\frac{\partial z}{\partial w}\frac{dw}{dt}$$

上定理还可推广到中间变量不是一元函数而是多元函数的情况: $z=f[\phi(x,y),\psi(x,y)]$.

链式法则

如果$u=\phi(x,y)$及$v=\psi(x,y)$都在点(x,y)可微且函数$z=f(u,v)$在对应点(u,v)可微，则

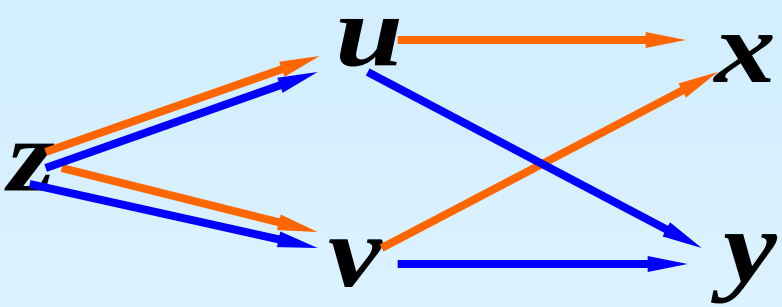
$$z=f[\phi(x,y),\psi(x,y)]$$

在对应点(x,y)可微,

且有

$$\frac{\partial z}{\partial x}=\frac{\partial z}{\partial u}\frac{\partial u}{\partial x}+\frac{\partial z}{\partial v}\frac{\partial v}{\partial x},\qquad \frac{\partial z}{\partial y}=\frac{\partial z}{\partial u}\frac{\partial u}{\partial y}+\frac{\partial z}{\partial v}\frac{\partial v}{\partial y}$$

链式法则如图示



$$\frac{\partial z}{\partial x}=\frac{\partial z}{\partial u}\cdot\frac{\partial u}{\partial x}+\frac{\partial z}{\partial v}\cdot\frac{\partial v}{\partial x},$$

$$\frac{\partial z}{\partial y}=\frac{\partial z}{\partial u}\cdot\frac{\partial u}{\partial y}+\frac{\partial z}{\partial v}\cdot\frac{\partial v}{\partial y}.$$

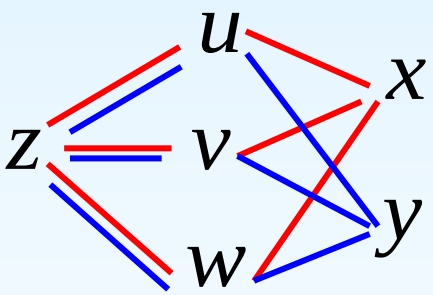
链式法则

类似地，

设$u=\phi(x,y)$、$v=\psi(x,y)$、$w=w(x,y)$都在点(x,y)可微，函数 $z=f(u,v,w)$在对应点可微，则称复合函数$z=f[\phi(x,y),\psi(x,y),w(x,y)]$在对应点$(x,y)$可微，且

$$\frac{\partial z}{\partial x}=\frac{\partial z}{\partial u}\frac{\partial u}{\partial x}+\frac{\partial z}{\partial v}\frac{\partial v}{\partial x}+\frac{\partial z}{\partial w}\frac{\partial w}{\partial x}$$

$$\frac{\partial z}{\partial y}=\frac{\partial z}{\partial u}\frac{\partial u}{\partial y}+\frac{\partial z}{\partial v}\frac{\partial v}{\partial y}+\frac{\partial z}{\partial w}\frac{\partial w}{\partial y}$$



推论2.1 n元函数的链式求导法则

$f\left(u_1,u_2,\ldots\ldots u_m\right)$在$\left(u_1,u_2,\ldots\ldots u_m\right)$可微,

$u_k\left(x_1,x_2,\ldots\ldots x_n\right),k=1,2,3\ldots\ldots,m$在

$\left(x_1,x_2,\ldots\ldots x_n\right)$可微

$$\Rightarrow \frac{\partial f}{\partial x_i}=\sum_{k=1}^{m}\frac{\partial f}{\partial u_k}\frac{\partial u_k}{\partial x_i},i=1,2,3,\ldots\ldots n.$$

n元函数的链式求导法则

特殊地 $z = f(u, x, y)$ 其中 $u = \phi(x, y)$

即 $z = f[\phi(x, y), x, y]$, 令 $v = x$, $w = y$,

$$\frac{\partial v}{\partial x} = 1, \quad \frac{\partial w}{\partial x} = 0, \quad \frac{\partial v}{\partial y} = 0, \quad \frac{\partial w}{\partial y} = 1.$$

$$\boxed{\frac{\partial z}{\partial x}} = \frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial x} + \boxed{\frac{\partial f}{\partial x}}, \qquad \boxed{\frac{\partial z}{\partial y}} = \frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial y} + \boxed{\frac{\partial f}{\partial y}}.$$

区别类似

两者的区别

把复合函数 $z = f[\phi(x, y), x, y]$ 中的 y 看作不变而对 x 的偏导数

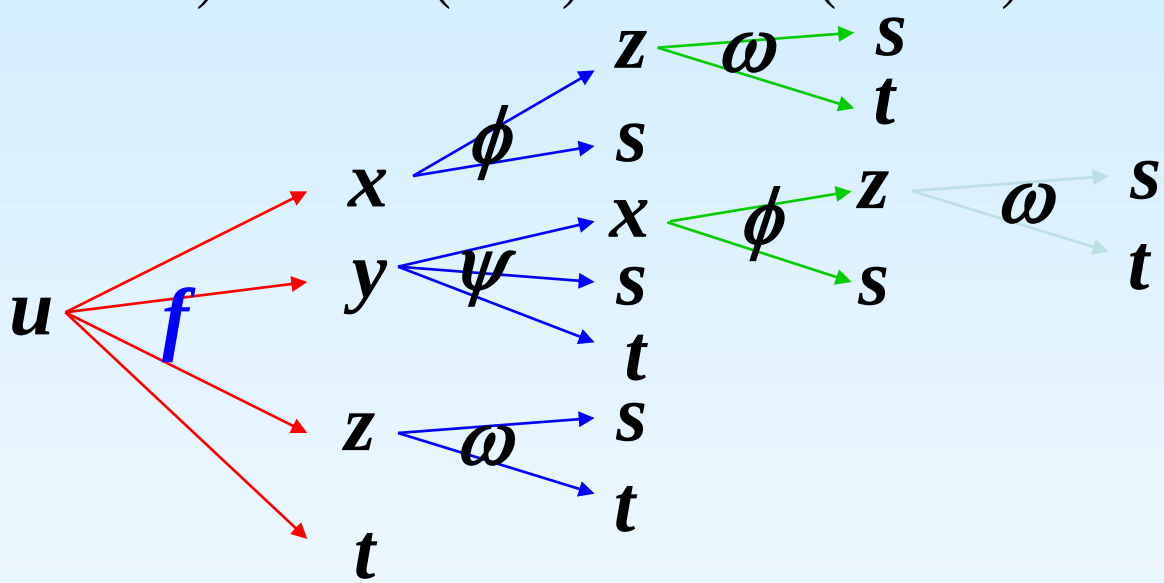
把 $z = f(u, x, y)$ 中的 u 及 y 看作不变而对 x 的偏导数

n元函数的链式求导法则

多重复合函数的求导：

$$u=f(x,y,z,t),x=\phi(z,s),y=\psi(x,s,t),z=\omega(s,t)$$

求 u_t,u_s



$$u_s=\frac{\partial f}{\partial x}\left[\frac{\partial\phi}{\partial z}\frac{\partial\omega}{\partial s}+\frac{\partial\phi}{\partial s}\right]+\frac{\partial f}{\partial y}\left[\frac{\partial\psi}{\partial x}\frac{\partial\phi}{\partial z}\frac{\partial\omega}{\partial s}+\frac{\partial\psi}{\partial x}\frac{\partial\phi}{\partial s}+\frac{\partial\psi}{\partial s}\right]+\frac{\partial f}{\partial z}\frac{\partial\omega}{\partial s}$$

$$u_t=\frac{\partial f}{\partial x}\frac{\partial\phi}{\partial z}\frac{\partial\omega}{\partial t}+\frac{\partial f}{\partial y}\left[\frac{\partial\psi}{\partial x}\frac{\partial\phi}{\partial z}\frac{\partial\omega}{\partial t}+\frac{\partial\psi}{\partial t}\right]+\frac{\partial f}{\partial z}\frac{\partial\omega}{\partial t}+\frac{\partial f}{\partial t}$$

注: 定理中条件的说明

如果仅仅是为了求 $z = f[\phi(x,y),\psi(x,y)]$

在点(x,y) 对x和y的偏导数，则只需有 $u=\phi(x,y)$

及$v=\psi(x,y)$都在点(x,y) 偏导数存在即可，

而使用链式法则，函数$z = f(u,v)$的可微性不

能省略.

n元函数的链式求导法则

例如 $z = f(x,y) = \begin{cases} \dfrac{x^2 y}{x^2+y^2}, & x^2+y^2 \neq 0 \\ 0, & x^2+y^2 = 0 \end{cases}$

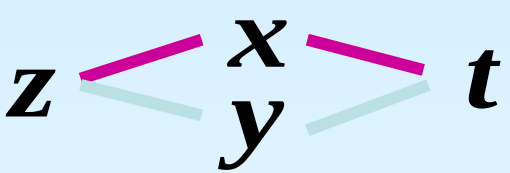
$f_x(0,0) = f_y(0,0) = 0$, $f(x,y)$ 在 $(0,0)$不可微.

如 $x = t, y = t$ 有 $z = F(t) = f(t,t) = \dfrac{t}{2}$, $\dfrac{dz}{dt} = \dfrac{1}{2}$,

$$\left.\frac{dz}{dt}\right|_{t=0} = \left.\frac{\partial z}{\partial x}\right|_{(0,0)} \left.\frac{dx}{dt}\right|_{t=0} + \left.\frac{\partial z}{\partial y}\right|_{(0,0)} \left.\frac{dy}{dt}\right|_{t=0} = 0.$$

例1　设$z=\arctan x+\ln y, x=\sin t, y=\tan t$,求$\dfrac{dz}{dt}$.

解:
$$\frac{dz}{dt}=\frac{\partial z}{\partial x}\frac{dx}{dt}+\frac{\partial z}{\partial y}\frac{dy}{dt}$$

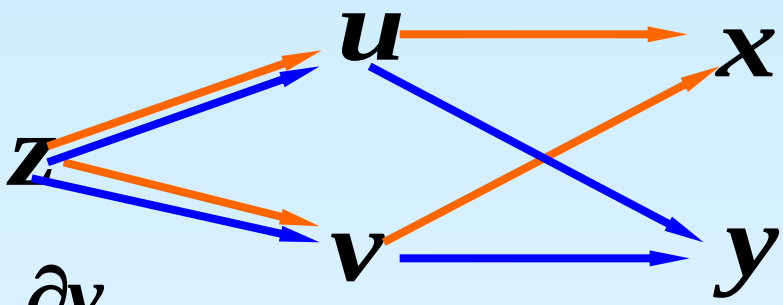


$$=\frac{1}{1+x^2}\cos t+\frac{1}{y}\sec^2 t$$

$$=\frac{\cos t}{1+\sin^2 t}+\frac{2}{\sin 2t}.$$

例 2 设$z = e^u \sin v$，而$u = xy$，$v = x + y$，

求 $\dfrac{\partial z}{\partial x}$ 和 $\dfrac{\partial z}{\partial y}$.



解: $$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x}$$

$$= e^u \sin v \cdot y + e^u \cos v \cdot 1 \ = e^u (y \sin v + \cos v),$$

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial y}$$

$$= e^u \sin v \cdot x + e^u \cos v \cdot 1 \ = e^u (x \sin v + \cos v).$$

例 3 设$z = uv + \sin t$，而$u = e^t$，$v = \cos t$，

求导数$\dfrac{dz}{dt}$.

$$\begin{tikzcd} & u \arrow[r] & t \\ z \arrow[ur] \arrow[r] \arrow[dr] & v \arrow[r] & t \\ & t & \end{tikzcd}$$

解:

$$\frac{dz}{dt} = \frac{\partial z}{\partial u} \cdot \frac{du}{dt} + \frac{\partial z}{\partial v} \cdot \frac{dv}{dt} + \frac{\partial z}{\partial t}$$

$$= ve^t - u\sin t + \cos t$$

$$= e^t \cos t - e^t \sin t + \cos t$$

$$= e^t(\cos t - \sin t) + \cos t.$$

二、全微分形式不变性

设函数$z=f(u,v)$可微，则有全微分$dz=\frac{\partial z}{\partial u}du+\frac{\partial z}{\partial v}dv$;

当$u=\phi(x,y)$、$v=\psi(x,y)$时，有$dz=\frac{\partial z}{\partial x}dx+\frac{\partial z}{\partial y}dy$.

$$=\left(\frac{\partial z}{\partial u}\cdot\frac{\partial u}{\partial x}+\frac{\partial z}{\partial v}\cdot\frac{\partial v}{\partial x}\right)dx+\left(\frac{\partial z}{\partial u}\cdot\frac{\partial u}{\partial y}+\frac{\partial z}{\partial v}\cdot\frac{\partial v}{\partial y}\right)dy$$

$$=\frac{\partial z}{\partial u}\left(\frac{\partial u}{\partial x}dx+\frac{\partial u}{\partial y}dy\right)+\frac{\partial z}{\partial v}\left(\frac{\partial v}{\partial x}dx+\frac{\partial v}{\partial y}dy\right)$$

$$=\frac{\partial z}{\partial u}du+\frac{\partial z}{\partial v}dv.$$

全微分形式不变形的实质

无论z是自变量u,v的函数或中间变量u,v的函数，它的全微分形式是一样的。

例 4　已知 $z = e^{xy}\sin(x+y)$，求 $\dfrac{\partial z}{\partial x}$ 和 $\dfrac{\partial z}{\partial y}$.

解： 令 $z = e^{u}\sin v,\ u = xy,\ v = x + y.$

$$dz = z_u du + z_v dv = e^{u}\sin v du + e^{u}\cos v dv$$

$$du = ydx + xdy,\ dv = dx + dy$$

$$dz = e^{xy}[y\sin(x+y) + \cos(x+y)]dx +$$

$$e^{xy}[x\sin(x+y) + \cos(x+y)]dy$$

$$\frac{\partial z}{\partial x} = e^{xy}[y\sin(x+y) + \cos(x+y)]$$

$$\frac{\partial z}{\partial y} = e^{xy}[x\sin(x+y) + \cos(x+y)]$$

三、小结

1、链式法则（特别要注意课中所讲的特殊情况）

2、全微分形式不变性（理解其实质）

思考题

设$z=f(u,v,x)$，而$u=\phi(x)$，$v=\psi(x)$，

则$\dfrac{dz}{dx}=\dfrac{\partial f}{\partial u}\dfrac{du}{dx}+\dfrac{\partial f}{\partial v}\dfrac{dv}{dx}+\dfrac{\partial f}{\partial x}$，

试问$\dfrac{dz}{dx}$与$\dfrac{\partial f}{\partial x}$是否相同？为什么？

思考题解答：

不相同.

等式左端的z是作为一个自变量 x的函数，

而等式右端最后一项f 是作为u, v, x的三元函数，

写出来为

$$\left.\frac{dz}{dx}\right|_x = \left.\frac{\partial f}{\partial u}\right|_{(u,v,x)} \cdot \left.\frac{du}{dx}\right|_x + \left.\frac{\partial f}{\partial v}\right|_{(u,v,x)} \cdot \left.\frac{dv}{dx}\right|_x + \left.\frac{\partial f}{\partial x}\right|_{(u,v,x)}.$$

作业 习题15-2

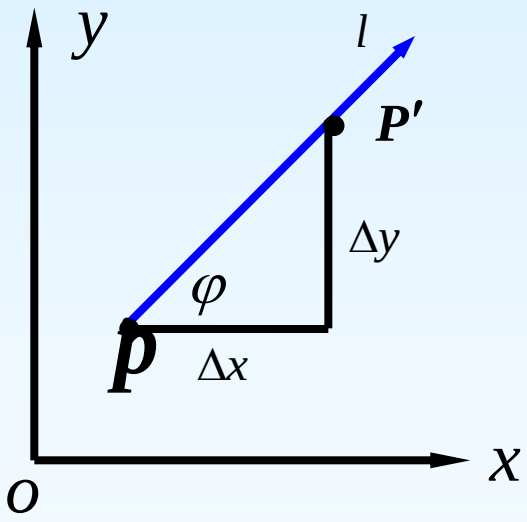
1

§15.3 方向导数和梯度

定义3.1 方向导数

讨论函数$z=f(x,y)$在一点P沿某一方向的变化率问题.

设函数$z=f(x,y)$在点$P(x,y)$的某一邻域$U(P)$内有定义，自点P引射线l. 设x轴正向到射线l的转角为φ,并设$P'(x+\Delta x,y+\Delta y)$为$l$上的另一点且$P'\in U(p)$.（如图）



考虑 $\dfrac{\Delta z}{\rho}$，如当 P'沿着 l 趋于P时，

$$\lim_{\rho\to 0}\frac{f(x+\Delta x, y+\Delta y)-f(x,y)}{\rho}$$

存在，称此极限为函数在点 P 沿方向 l 的方向导数。

记为

$$\frac{\partial f}{\partial l}=\lim_{\rho\to 0}\frac{f(x+\Delta x, y+\Delta y)-f(x,y)}{\rho}.$$

依定义，若函数 $f(x,y)$ 在点 p 存在偏导数，函数 $f(x,y)$ 在点 P 沿着 x 轴正向 $\vec{e}_1=\{1,0\}$、y 轴正向 $\vec{e}_2=\{0,1\}$ 的方向导数分别为 f_x, f_y；沿着 x 轴负向、y 轴负向的方向导数是 $-f_x, -f_y$.

定理 3.1： 如果函数 $z=f(x,y)$ 在点 $P(x,y)$ 是可微的，那么函数在该点沿任意方向 l 的方向导数都存在，且有

$$\frac{\partial f}{\partial l}=\frac{\partial f}{\partial x}\cos\varphi+\frac{\partial f}{\partial y}\sin\varphi,$$

其中 φ 为 x 轴正向到方向 l 的转角.

证明：由于函数可微，则增量可表示为

$$f(x+\Delta x, y+\Delta y)-f(x,y)=\frac{\partial f}{\partial x}\Delta x+\frac{\partial f}{\partial y}\Delta y+o(\rho)$$

两边同除以 ρ，得到

$$\frac{f(x+\Delta x, y+\Delta y)-f(x,y)}{\rho}=\frac{\partial f}{\partial x}\cdot\underbrace{\frac{\Delta x}{\rho}}_{\cos\varphi}+\frac{\partial f}{\partial y}\cdot\underbrace{\frac{\Delta y}{\rho}}_{\sin\varphi}+\frac{o(\rho)}{\rho}$$

故有方向导数：

$$\frac{\partial f}{\partial l}=\lim_{\rho\to 0}\frac{f(x+\Delta x, y+\Delta y)-f(x,y)}{\rho}=\frac{\partial f}{\partial x}\cos\varphi+\frac{\partial f}{\partial y}\sin\varphi.$$

二、梯度的概念(gradient)

问题:函数在点 P 沿哪一方向增加的速度最快?

定义 3.2：设函数$z=f(x,y)$在平面区域 D 内具有所有变量的偏导数，则对于每一点$P(x,y)\in D$，都可定出一个向量$\frac{\partial f}{\partial x}\vec{i}+\frac{\partial f}{\partial y}\vec{j}$，这向量称为函数$z=f(x,y)$在点$P(x,y)$的梯度，记为

$$gradf(x,y)=\frac{\partial f}{\partial x}\vec{i}+\frac{\partial f}{\partial y}\vec{j}.$$

梯度的概念

设$\vec{e}=\cos\varphi\vec{i}+\sin\varphi\vec{j}$是方向$\vec{l}$上的单位向量，

由方向导数公式知

$$\frac{\partial f}{\partial l}=\frac{\partial f}{\partial x}\cos\varphi+\frac{\partial f}{\partial y}\sin\varphi=(\frac{\partial f}{\partial x},\frac{\partial f}{\partial y})\cdot(\cos\varphi,\sin\varphi)$$

$$=gradf(x,y)\cdot\vec{e}=|gradf(x,y)|\cos\theta,$$

其中$\theta=(gradf(x,y),\vec{e})$

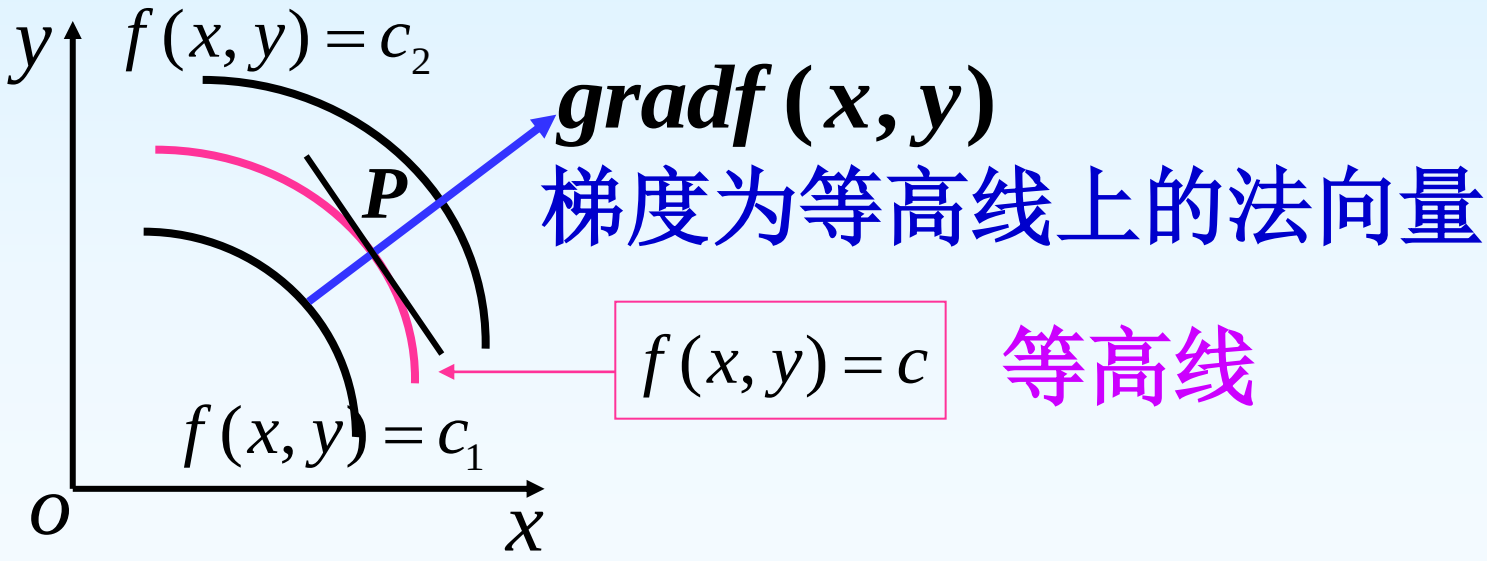
当 $\cos(gradf(x,y),\vec{e})=1$时，$\frac{\partial f}{\partial l}$有最大值.

梯度的概念

在几何上 $z = f(x, y)$ 表示一个曲面

曲面被平面 $z = c$ 所截得 $\begin{cases} z = f(x, y) \\ z = c \end{cases}$,

所得曲线在xoy面上投影如图



例 1：求函数 $f(x,y)=x^2-xy+y^2$ 在点（1，1）沿与 x 轴方向夹角为 α 的方向射线 $\vec{l}$ 的方向导数.并问在怎样的方向上此方向导 数有

（1）最大值；　（2）最小值；　（3）等于零？

解：由方向导数的计算公式知

$$\left.\frac{\partial f}{\partial l}\right|_{(1,1)}=f_x(1,1)\cos\alpha+f_y(1,1)\sin\alpha$$

$$=(2x-y)\big|_{(1,1)}\cos\alpha+(2y-x)\big|_{(1,1)}\sin\alpha,$$

$$=\cos\alpha+\sin\alpha=\sqrt{2}\sin(\alpha+\frac{\pi}{4}),$$

故（1）当$\alpha=\dfrac{\pi}{4}$时，方向导数达到最大值$\sqrt{2}$；

（2）当$\alpha=\dfrac{5\pi}{4}$时，方向导数达到最小值$-\sqrt{2}$；

（3）当$\alpha=\dfrac{3\pi}{4}$和$\alpha=\dfrac{7\pi}{4}$时，方向导数等于 0.

推广可得三元函数方向导数的定义

对于三元函数$u=f(x,y,z)$，它在空间一点$P(x,y,z)$沿着方向l的方向导数，可定义为：

$$\frac{\partial f}{\partial l}=\lim_{\rho\to 0}\frac{f(x+\Delta x,y+\Delta y,z+\Delta z)-f(x,y,z)}{\rho},$$

（其中$\rho=\sqrt{(\Delta x)^2+(\Delta y)^2+(\Delta z)^2}$）

三元函数的方向导数

设方向l的方向角为α,β,γ

$$\Delta x=\rho\cos\alpha,\quad \Delta y=\rho\cos\beta,\quad \Delta z=\rho\cos\gamma,$$

同理：当函数在此点可微时，那么函数在该点沿任意方向l的方向导数都存在，且有

$$\frac{\partial f}{\partial l}=\frac{\partial f}{\partial x}\cos\alpha+\frac{\partial f}{\partial y}\cos\beta+\frac{\partial f}{\partial z}\cos\gamma.$$

三元函数的梯度

梯度的概念可以推广到三元函数

$$gradf(x,y,z)=\frac{\partial f}{\partial x}\vec{i}+\frac{\partial f}{\partial y}\vec{j}+\frac{\partial f}{\partial z}\vec{k}.$$

类似于二元函数，此梯度也是一个向量，其方向与取得最大方向导数的方向一致，其模为方向导数的最大值.

例 2：求函数 $u = x^2 + 2y^2 + 3z^2 + 3x - 2y$ 在点 (1,1,2)处的梯度，并问在 哪些点处梯度为零？

解：由梯度计算公式得

$$gradu(x,y,z) = \frac{\partial u}{\partial x}\vec{i} + \frac{\partial u}{\partial y}\vec{j} + \frac{\partial u}{\partial z}\vec{k}$$

$$= (2x+3)\vec{i} + (4y-2)\vec{j} + 6z\vec{k},$$

故 $gradu(1,1,2) = 5\vec{i} + 2\vec{j} + 12\vec{k}.$

在 $P_0(-\frac{3}{2},\frac{1}{2},0)$ 处梯度为 0.

三、小结

1、方向导数的概念

（注意方向导数与一般所说偏导数的区别）

2、梯度的概念

（注意梯度是一个向量）

3、方向导数与梯度的关系

梯度的方向就是函数 $f(x,y)$ 在这点增长最快的方向.

思考题

1. 讨论函数$z = f(x,y) = \sqrt{x^2 + y^2}$在$(0,0)$点处的偏导数是否存在？方向导数是否存在？

2. 考虑下面各项之间的关系

f可微　　f连续　　f_x, f_y, f_z存在

沿任意方向的方向导数存在

$$\frac{\partial f}{\partial(\pm i)}, \frac{\partial f}{\partial(\pm j)}, \frac{\partial f}{\partial(\pm k)}\text{存在}$$

思考题解答

$$\left.\frac{\partial z}{\partial x}\right|_{(0,0)} = \lim_{\Delta x \to 0} \frac{f(\Delta x, 0) - f(0,0)}{\Delta x} = \lim_{\Delta x \to 0} \frac{|\Delta x|}{\Delta x}.$$

同理：$\left.\frac{\partial z}{\partial y}\right|_{(0,0)} = \lim_{\Delta y \to 0} \frac{|\Delta y|}{\Delta y}$

故两个偏导数均不存在.

思考题解答

沿任意方向$\vec{l}=\{x,y,z\}$的方向导数,

$$\left.\frac{\partial z}{\partial l}\right|_{(0,0)}=\lim_{\rho\to 0}\frac{f(\Delta x,\Delta y)-f(0,0)}{\rho}$$

$$=\lim_{\rho\to 0}\frac{\sqrt{(\Delta x)^2+(\Delta y)^2}}{\sqrt{(\Delta x)^2+(\Delta y)^2}}=1$$

故沿任意方向的方向导数均存在且相等.

作业 习题15-3

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§15.4 高阶偏导

一、高阶偏导数

函数$z=f(x,y)$的二阶偏导数为

$$\frac{\partial}{\partial x}\left(\frac{\partial z}{\partial x}\right)=\frac{\partial^2 z}{\partial x^2}=f_{xx}(x,y),\quad \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial y}\right)=\frac{\partial^2 z}{\partial y^2}=f_{yy}(x,y)$$

纯偏导

$$\frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right)=\frac{\partial^2 z}{\partial x\partial y}=f_{xy}(x,y),\quad \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial y}\right)=\frac{\partial^2 z}{\partial y\partial x}=f_{yx}(x,y)$$

混合偏导

定义：二阶及二阶以上的偏导数统称为高阶偏导数.

例 1：验证函数$u(x,y)=\ln\sqrt{x^2+y^2}$满足拉普拉

斯方程 $\dfrac{\partial^2 u}{\partial x^2}+\dfrac{\partial^2 u}{\partial y^2}=0.$

解： $\because \ln\sqrt{x^2+y^2}=\dfrac{1}{2}\ln(x^2+y^2),$

$$\therefore \frac{\partial u}{\partial x}=\frac{x}{x^2+y^2},\quad \frac{\partial u}{\partial y}=\frac{y}{x^2+y^2},$$

高阶偏导数

$$\therefore \frac{\partial^2 u}{\partial x^2}=\frac{(x^2+y^2)-x\cdot 2x}{(x^2+y^2)^2}=\frac{y^2-x^2}{(x^2+y^2)^2},$$

$$\frac{\partial^2 u}{\partial y^2}=\frac{(x^2+y^2)-y\cdot 2y}{(x^2+y^2)^2}=\frac{x^2-y^2}{(x^2+y^2)^2}.$$

$$\therefore \frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}=\frac{y^2-x^2}{(x^2+y^2)^2}+\frac{x^2-y^2}{(x^2+y^2)^2}=0.$$

例2: 设$z = x^3y^2 - 3xy^3 - xy + 1$, :求$\dfrac{\partial^2 z}{\partial x^2}$、$z_{yx}$、$z_{xy}$、$\dfrac{\partial^2 z}{\partial y^2}$及$\dfrac{\partial^3 z}{\partial x^3}$.

解: $\dfrac{\partial z}{\partial x} = 3x^2y^2 - 3y^3 - y$, $\dfrac{\partial z}{\partial y} = 2x^3y - 9xy^2 - x$;

$$\frac{\partial^2 z}{\partial x^2} = 6xy^2,\ \frac{\partial^3 z}{\partial x^3} = 6y^2,\ \frac{\partial^2 z}{\partial y^2} = 2x^3 - 18xy;$$

$$z_{xy} = 6x^2y - 9y^2 - 1, \quad z_{yx} = 6x^2y - 9y^2 - 1.$$

例 3 ： $f(x,y)$, $x(s,t)$, $y(s,t)$都具有二阶连续偏导数.

且$u = f(x(s,t), y(s,t))$,求:$\dfrac{\partial^2 u}{\partial s^2}$

解:
$$\frac{\partial u}{\partial s} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial s},$$

$$\frac{\partial^2 u}{\partial s^2} = \frac{\partial}{\partial s}\left(\frac{\partial f}{\partial x}\frac{\partial x}{\partial s}\right) + \frac{\partial}{\partial s}\left(\frac{\partial f}{\partial y}\frac{\partial y}{\partial s}\right),$$

$$= \frac{\partial}{\partial s}\left(\frac{\partial f}{\partial x}\right)\frac{\partial x}{\partial s} + \frac{\partial f}{\partial x}\frac{\partial^2 x}{\partial s^2} + \frac{\partial}{\partial s}\left(\frac{\partial f}{\partial y}\right)\frac{\partial y}{\partial s} + \frac{\partial f}{\partial y}\frac{\partial^2 y}{\partial s^2}$$

$$=\frac{\partial}{\partial s}\left(\frac{\partial f}{\partial x}\right)\frac{\partial x}{\partial s}+\frac{\partial f}{\partial x}\frac{\partial^2 x}{\partial s^2}+\frac{\partial}{\partial s}\left(\frac{\partial f}{\partial y}\right)\frac{\partial y}{\partial s}+\frac{\partial f}{\partial y}\frac{\partial^2 y}{\partial s^2}$$

$$\boxed{\frac{\partial^2 f}{\partial x^2}\frac{\partial x}{\partial s}+\frac{\partial^2 f}{\partial y\partial x}\frac{\partial y}{\partial s}}$$

$$\boxed{\frac{\partial^2 f}{\partial x\partial y}\frac{\partial x}{\partial s}+\frac{\partial^2 f}{\partial y^2}\frac{\partial y}{\partial s}}$$

$$=\frac{\partial^2 f}{\partial x^2}\left(\frac{\partial x}{\partial s}\right)^2+2\frac{\partial^2 f}{\partial x\partial y}\frac{\partial x}{\partial s}\frac{\partial y}{\partial s}+\frac{\partial^2 f}{\partial y^2}\left(\frac{\partial y}{\partial s}\right)^2$$

$$+\frac{\partial f}{\partial x}\frac{\partial^2 x}{\partial s^2}+\frac{\partial f}{\partial y}\frac{\partial^2 y}{\partial s^2}$$

其他二阶导数练习

问题：

混合偏导数都相等吗？具备怎样的条件才相等？

例4：$$f(x,y)=\begin{cases} xy\dfrac{x^2-y^2}{x^2+y^2} & x^2+y^2\neq 0, \\ 0\quad, & x^2+y^2=0, \end{cases}$$

$$f_x(x,y)=\begin{cases} \dfrac{y(x^4+4x^2y^2-y^4)}{(x^2+y^2)^2} & x^2+y^2\neq 0, \\ 0\quad, & x^2+y^2=0, \end{cases}$$

$$f_y(x,y)=\begin{cases}\dfrac{x(x^4-4x^2y^2-y^4)}{(x^2+y^2)^2}, & x^2+y^2\neq 0,\\ 0, & x^2+y^2=0,\end{cases}$$

$$f_{xy}(0,0)=-1,\quad f_{yx}(0,0)=1.$$

定理 4.1：如果函数$z=f(x,y)$的两个二阶混合偏导数f_{xy}及f_{yx}在区域 D 内连续，那么在该区域内这两个二阶混合偏导数必相等.

作业 习题15-4

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